pyxtdpthresh 0.0.1: validation against Stata 17

Largest deviation by block, by case and by object

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1 Rule and method

Every number the package reports must equal the one computed by Stata 17 to four decimals: the absolute difference is at most $5e-05$ *or* the relative difference is at most $1e-06$ (the relative branch keeps the rule meaningful for large values such as the objective function). The *margin* of a value is $\min(|\Delta|/5e-05,\ (|\Delta|/|x|)/1e-06)$; it is at most 1 when the value agrees, and the tables report the largest margin of each block.

The references are produced by the real commands (xthenreg (Seo, Kim and Kim 2019) and xtdpthresh 0.9.38) run in batch with Stata 17.0 MP; each object is exported at 15 significant digits and frozen in `tests/golden/`. The test suite compares the package with the frozen values, and the deviations in this report are recorded by those very comparisons (4\,764 comparisons of matrices or numbers in the golden tests, plus the objects of the help-file examples). The random numbers of Stata are reproduced exactly, so the seeded bootstraps draw the weights of the command and their p-values and confidence sets are compared directly.

2 Summary by block

Block	Cases	Objects	Values	Largest $\lvert\Delta\rvert$	Margin
Random numbers (Stata's Mersenne Twister)	5	1	10 500	1.3×10^{-15}	2.7e-11
Matrix routines (invsym cholinv syminv)	22	5	1 210	4.6×10^{5}	0.097
Quantiles and threshold grids	5	9	1 930	3.7×10^{-13}	3.7e-09
Data layer: xtset lags missing values	3	2	113 223	0	0
Time-series operators	19	19	4 838	5.6×10^{-17}	1.1e-12
xthenreg: data and instruments	12	14	44 849	2.3×10^{-13}	5.1e-10
xthenreg: sup-Wald statistic and linearity test	10	4	2 309	6.3×10^{-5}	0.00019
xthenreg: covariance and confidence intervals	24	4	2 324	1.4×10^{-12}	2.8e-08
xthenreg: complete result	32	6	2 916	4.1×10^{-11}	8.1e-07
xthenreg: bandwidth multiplier h_0() through the public function	2	3	260	3.7×10^{-13}	7.3e-09
xtdpthresh: instruments and fixed-threshold GMM	28	1	962 700	5.1×10^{-15}	1e-10
xtdpthresh: threshold search and final estimate	29	6	1 537	9.5×10^{-10}	1.6e-06
xtdpthresh: point estimates on the full panel	14	4	1 064	1.1×10^{-9}	2.1e-05
xtdpthresh: deviations time effects unbalanced panels history	41	7	1 508 791	7.6×10^{-9}	0.00011
xtdpthresh: variances and diagnostics	45	32	7 244	1.1×10^{-9}	2.1e-05
xtdpthresh: predict	12	6	66 718	3.3×10^{-13}	6.6e-09
xtdpthresh: bootstraps confidence set tests	56	29	14 538	8.9×10^{-8}	8.2e-05
xtdpthresh: unbalanced panel of the help file (no bootstrap)	2	4	184	6.6×10^{-14}	1.3e-09
xtdpthresh: hostile inputs	4	1	24	5.9×10^{-8}	0.0012
Simulated panels of the Seo-Shin design	4	1	20	4.5×10^{-9}	8.9e-05
xtdpthresh: the examples of the help file (default options)	8	72	503	2.1×10^{-7}	0.0014
All blocks	**377**		**2 747 682**	4.6×10^{5}	**0.097**

Cases are the distinct designs or samples of a block; *Objects* are the distinct names compared (for example `e(b)`); *Values* counts every number compared, a matrix counting for all its elements. The absolute deviation is large only where the values are (the matrix routines are tested on very large and very small matrices);

the margin says whether the rule holds. Integer results (counts of equations, instruments, return codes of the refusals, sets of rows) are compared for exact equality and are not listed in the deviations: the tests fail on the first difference.

3 The examples of the help file

The eight commands of the help file of `xtdpthresh` were run as printed, with the default options of Stata (`boot(299)`, `gridci(100)`, `grid(100)`), on Hansen's investment panel without the five extreme firms. The tables of the values, Python against Stata, are in `docs/help_examples.md`; the largest deviation of each kind of object is below.

Example	Objects	Compared	Largest $\lvert\Delta\rvert$
Point estimates and diagnostics	coefficient	9	1.3×10^{-14}
	std. error	9	1.2×10^{-14}
	scalar	30	9.2×10^{-14}
	matrix	5	1.6×10^{-14}
Full inference and the test of Hansen's threshold	coefficient	9	1.3×10^{-14}
	std. error	9	1.2×10^{-14}
	scalar	45	2.3×10^{-11}
	matrix	7	4.9×10^{-9}
Instruments from lag 2	coefficient	9	1.8×10^{-9}
	std. error	9	5.9×10^{-9}
	scalar	45	1.0×10^{-9}
	matrix	7	2.1×10^{-7}
Kink model	coefficient	5	3.0×10^{-12}
	std. error	5	8.0×10^{-11}
	scalar	45	2.7×10^{-12}
	matrix	7	6.4×10^{-10}
Unbalanced panel, first differences	coefficient	9	3.3×10^{-14}
	std. error	9	6.7×10^{-13}
	scalar	45	1.6×10^{-13}
	matrix	7	3.3×10^{-10}
Unbalanced panel, forward orthogonal deviations	coefficient	9	3.8×10^{-10}
	std. error	9	7.5×10^{-10}
	scalar	45	9.2×10^{-11}
	matrix	7	1.7×10^{-8}
External instrument	coefficient	9	1.2×10^{-9}
	std. error	9	3.1×10^{-9}
	scalar	30	1.7×10^{-9}
	matrix	5	1.0×10^{-7}
Postestimation: residuals and regime	coefficient	9	1.3×10^{-14}
	std. error	9	1.2×10^{-14}
	scalar	30	9.2×10^{-14}
	matrix	5	1.6×10^{-14}
	predict	2	2.7×10^{-14}

4 Largest deviation by object

The objects are named as the tests name them (`e(b)`, `e(V)`, `e(ci_grid)`…). A block appears here when it compares several named objects over several cases; the other blocks compare one kind of value, or one object per case, and are described by the table of their cases in the next section.

Matrix routines (invsym, cholinv, syminv)

Object	Cases	Values	Largest $\lvert\Delta\rvert$
`cholinv`	14	384	4.6×10^{5}
`invsym`	21	439	1.2×10^{-1}

Object	Cases	Values	Largest $\lvert\Delta\rvert$
psd	22	22	0
syminv	13	343	4.6×10^{5}
syminv_ok	22	22	0

Quantiles and threshold grids

Object	Cases	Values	Largest $\lvert\Delta\rvert$
adm	5	500	0
minreg	5	10	0
mm	5	140	4.6×10^{-14}
qgrid	5	130	7.8×10^{-14}
qgrid1	5	5	1.8×10^{-15}
rangen	5	500	5.0×10^{-14}
rangen1	5	5	1.8×10^{-15}
rangen_x	5	500	5.7×10^{-14}
t7	5	140	3.7×10^{-13}

Data layer: xtset, lags, missing values

Object	Cases	Values	Largest $\lvert\Delta\rvert$
balance	3	3	0
rows	3	113 220	0

xthenreg: data and instruments

Object	Cases	Values	Largest $\lvert\Delta\rvert$
Z.bilinear	12	12	1.1×10^{-13}
Z.cols	12	6 668	2.3×10^{-13}
Z.rows	12	1 140	1.1×10^{-13}
moment	12	453	0
q.bilinear	12	12	0
q.cols	12	105	0
q.rows	12	6 668	0
scal	12	60	0
x.bilinear	12	12	0
x.cols	12	105	0
x.rows	12	22 829	0
y.bilinear	12	12	0
y.cols	12	105	0
y.rows	12	6 668	0

xthenreg: sup-Wald statistic and linearity test

Object	Cases	Values	Largest $\lvert\Delta\rvert$
boot_sup	9	360	6.3×10^{-5}
cov_wald	9	1 900	1.4×10^{-6}
sup_wald	9	9	3.5×10^{-5}
value	1	40	2.8×10^{-14}

xthenreg: covariance and confidence intervals

Object	Cases	Values	Largest $\lvert\Delta\rvert$
CI	16	272	1.4×10^{-12}

Object	Cases	Values	Largest $\lvert\Delta\rvert$
V	16	1 232	1.6×10^{-13}
b	16	136	8.8×10^{-13}
value	8	684	6.9×10^{-18}

xthenreg: complete result

Object	Cases	Values	Largest $\lvert\Delta\rvert$
CI	32	518	2.4×10^{-11}
N	32	36	0
T	32	36	0
V	32	2 031	4.1×10^{-11}
b	32	259	6.4×10^{-12}
boots_p	32	36	0

xthenreg: bandwidth multiplier h_0() through the public function

Object	Cases	Values	Largest $\lvert\Delta\rvert$
CI	2	40	3.7×10^{-13}
V	2	200	5.1×10^{-14}
b	2	20	1.7×10^{-13}

xtdpthresh: threshold search and final estimate

Object	Cases	Values	Largest $\lvert\Delta\rvert$
e(V_cond)	29	1 208	5.2×10^{-13}
e(b)	29	184	1.7×10^{-12}
e(obj)	29	29	9.5×10^{-10}
first-stage threshold	29	29	3.3×10^{-16}
gamma	29	29	5.0×10^{-16}
q bounds	29	58	2.2×10^{-16}

xtdpthresh: point estimates on the full panel

Object	Cases	Values	Largest $\lvert\Delta\rvert$
e(V_cond)	14	925	1.6×10^{-12}
e(b)	14	111	2.5×10^{-12}
e(gamma)	14	14	5.0×10^{-16}
e(obj)	14	14	1.1×10^{-9}

xtdpthresh: deviations, time effects, unbalanced panels, history

Object	Cases	Values	Largest $\lvert\Delta\rvert$
X at the threshold	41	160 566	2.5×10^{-14}
e(V_cond)	41	1 855	1.6×10^{-10}
e(b)	41	546	1.9×10^{-10}
e(obj)	41	82	7.6×10^{-9}
first-stage threshold	41	41	4.4×10^{-16}
gamma	41	41	4.9×10^{-16}
value	41	1 345 660	1.5×10^{-13}

xtdpthresh: variances and diagnostics

Object	Cases	Values	Largest $\lvert\Delta\rvert$
e(N)	45	45	0
e(N_clust)	45	45	0
e(N_switch)	45	45	0
e(N_units)	45	45	0
e(V)	45	2 217	8.2×10^{-10}
e(V_cond)	45	2 217	2.4×10^{-10}
e(V_upper)	41	393	2.7×10^{-12}
e(ar1)	45	90	4.1×10^{-11}
e(ar1_T1)	45	90	1.9×10^{-10}
e(ar1_TT)	45	90	1.4×10^{-10}
e(ar1_b0)	45	90	2.2×10^{-11}
e(ar1_cond)	45	90	1.6×10^{-11}
e(ar1_p)	45	90	2.8×10^{-13}
e(ar2)	45	90	1.6×10^{-11}
e(ar2_T1)	45	90	9.1×10^{-12}
e(ar2_TT)	45	90	1.5×10^{-11}
e(ar2_b0)	45	90	2.9×10^{-12}
e(ar2_cond)	45	90	1.1×10^{-11}
e(ar2_p)	45	90	1.2×10^{-11}
e(ar_joint)	45	45	0
e(b)	45	309	2.9×10^{-11}
e(b_upper)	41	123	2.7×10^{-12}
e(estimator_twostep)	45	45	0
e(gamma)	45	90	4.4×10^{-16}
e(gamma_bw)	38	76	5.1×10^{-16}
e(hansen)	43	86	1.1×10^{-9}
e(hansen_df)	43	86	0
e(hansen_p)	43	86	5.7×10^{-11}
e(joint_vce)	45	45	0
e(obj)	45	45	1.1×10^{-9}
e(q_nvals_bw)	38	76	0
e(vce_applied)	45	45	0

xtdpthresh: predict

Object	Cases	Values	Largest $\lvert\Delta\rvert$
predict ar_residuals	11	13 098	3.3×10^{-13}
predict regime	11	16 973	0
predict residuals	11	13 144	3.3×10^{-13}
predict residuals if	11	9 659	3.2×10^{-13}
predict xb	11	13 144	3.3×10^{-13}
xb + residuals	1	700	6.9×10^{-18}

xtdpthresh: bootstraps, confidence set, tests

Object	Cases	Values	Largest $\lvert\Delta\rvert$
boot_linearity_valid	55	55	0
continuity valid	9	9	0
e(b_bootci)	10	134	7.8×10^{-13}
e(boot_coef_B)	10	10	0
e(boot_coef_attempted)	10	10	0
e(boot_coef_failed)	10	10	0
e(boot_coef_twostep)	10	10	0
e(boot_grid_stage1)	10	10	0
e(boot_grid_stage2)	10	10	0
e(boundary_warn)	56	56	0
e(ci_empty)	52	104	0
e(ci_grid)	56	13 002	8.9×10^{-8}
e(ci_incomplete)	56	112	0
e(ci_nseg)	52	104	0
e(ci_segments)	52	348	5.0×10^{-16}

Object	Cases	Values	Largest $\lvert\Delta\rvert$
e(ci_unresolved)	56	56	0
e(citest_D)	8	8	2.0×10^{-9}
e(citest_accept)	8	8	0
e(citest_crit)	8	8	5.0×10^{-9}
e(citest_draws)	9	9	0
e(citest_gamma)	9	9	0
e(citest_p)	8	8	0
e(citest_status)	9	9	0
e(gamma)	56	56	4.4×10^{-16}
e(gamma_hi)	52	104	5.0×10^{-16}
e(gamma_lo)	52	104	2.8×10^{-16}
e(pval_cont)	9	9	0
e(pval_lin)	55	110	0
e(rseed)	56	56	0

xtdpthresh: unbalanced panel of the help file (no bootstrap)

Object	Cases	Values	Largest $\lvert\Delta\rvert$
e(V_cond)	2	162	1.9×10^{-15}
e(b)	2	18	6.7×10^{-15}
e(gamma)	2	2	3.9×10^{-16}
e(obj)	2	2	6.6×10^{-14}

5 Largest deviation by case

Random numbers (Stata's Mersenne Twister)

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
0	1	2 100	value	1.3×10^{-15}
1	1	2 100	value	1.3×10^{-15}
123	1	2 100	value	1.3×10^{-15}
12345	1	2 100	value	1.3×10^{-15}
2147483647	1	2 100	value	1.1×10^{-15}

Matrix routines (invsym, cholinv, syminv)

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
asym	5	29	syminv	4.4×10^{-15}
cond10	4	52	cholinv	4.6×10^{2}
cond11	4	52	cholinv	2.8×10^{3}
cond6	5	77	syminv	6.9×10^{-6}
cond8	5	77	cholinv	3.3×10^{-2}
diag_wide	5	14	invsym	0
diag_zero	3	11	invsym	0
gram	5	77	invsym	4.3×10^{-17}
gram_sing	3	18	invsym	2.8×10^{-17}
huge	5	50	syminv	4.0×10^{-27}
indef	3	6	invsym	0
indef3	3	11	invsym	3.3×10^{-16}
mixed	4	20	syminv	1.6×10^{1}
nan	2	2	syminv_ok	0
negdef	3	6	invsym	0
ones	3	6	invsym	0
spd1	5	5	invsym	4.4×10^{-16}
spd12	5	434	invsym	3.9×10^{-14}
spd2	5	14	invsym	3.1×10^{-15}
spd4	5	50	syminv	5.1×10^{-15}

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
spd7	5	149	invsym	7.7×10^{-12}
tiny	5	50	cholinv	4.6×10^{5}

Quantiles and threshold grids

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
n300	9	390	qgrid	7.8×10^{-14}
skew1000	9	388	rangen_x	5.3×10^{-14}
small25	9	392	rangen	7.1×10^{-15}
ties10	9	370	rangen	6.2×10^{-15}
uni4520	9	390	t7	3.7×10^{-13}

Data layer: xtset, lags, missing values

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
full	2	40 681	balance	0
unbal	2	35 593	balance	0
weak	2	36 949	balance	0

Time-series operators

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
D.cashflow	1	300	D.cashflow	5.6×10^{-17}
D2.cashflow	1	222	D2.cashflow	5.6×10^{-17}
DF.cashflow	1	300	DF.cashflow	5.6×10^{-17}
DL.cashflow	1	222	DL.cashflow	5.6×10^{-17}
F.cashflow	1	300	F.cashflow	0
F2.cashflow	1	278	F2.cashflow	0
FD.cashflow	1	300	FD.cashflow	5.6×10^{-17}
L.D2.cashflow	1	153	L.D2.cashflow	5.6×10^{-17}
L.cashflow	1	300	L.cashflow	0
L2.cashflow	1	278	L2.cashflow	0
L2D.cashflow	1	205	L2D.cashflow	5.6×10^{-17}
L3.cashflow	1	257	L3.cashflow	0
LD.cashflow	1	222	LD.cashflow	5.6×10^{-17}
LF.cashflow	1	386	LF.cashflow	0
LS2.cashflow	1	205	LS2.cashflow	5.6×10^{-17}
S2.cashflow	1	278	S2.cashflow	5.6×10^{-17}
S2D.cashflow	1	153	S2D.cashflow	5.6×10^{-17}
S3.cashflow	1	257	S3.cashflow	5.6×10^{-17}
SS.cashflow	1	222	SS.cashflow	5.6×10^{-17}

xthenreg: data and instruments

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
c1	14	4 113	scal	0
c2	14	4 113	scal	0
c3	14	4 093	Z.cols	1.4×10^{-14}
c4	14	4 124	Z.cols	2.3×10^{-13}
c5	14	3 486	Z.cols	1.4×10^{-14}
c6	14	4 110	Z.cols	7.1×10^{-15}
f1	14	3 708	Z.cols	2.8×10^{-14}
f2	14	3 866	Z.cols	1.7×10^{-13}
m1	14	3 169	Z.cols	7.1×10^{-15}
m2	14	3 150	Z.cols	7.1×10^{-15}
w1	14	3 465	Z.cols	2.8×10^{-14}

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
w2	14	3 452	Z.cols	1.1×10^{-13}

xthenreg: sup-Wald statistic and linearity test

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
injected_draws_reproduce_the_same_statistic_as_the_generator	1	40	value	2.8×10^{-14}
l1	3	541	cov_wald	1.8×10^{-9}
l2	3	121	sup_wald	2.3×10^{-7}
l3	3	361	cov_wald	1.4×10^{-6}
l4	3	121	sup_wald	2.6×10^{-11}
l5	3	541	cov_wald	1.0×10^{-9}
p1	3	121	boot_sup	2.9×10^{-12}
p2	3	221	boot_sup	2.5×10^{-12}
p3	3	121	boot_sup	6.3×10^{-5}
p4	3	121	sup_wald	9.8×10^{-8}

xthenreg: covariance and confidence intervals

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
c1	1	110	value	6.9×10^{-18}
c1-fast	3	130	CI	3.1×10^{-13}
c1-literal	3	130	CI	1.3×10^{-13}
c2	1	42	value	2.7×10^{-20}
c2-fast	3	54	CI	2.2×10^{-14}
c2-literal	3	54	CI	1.6×10^{-14}
c3	1	72	value	1.1×10^{-19}
c3-fast	3	88	CI	1.4×10^{-12}
c3-literal	3	88	CI	8.9×10^{-13}
c4	1	110	value	4.3×10^{-19}
c4-fast	3	130	CI	4.3×10^{-14}
c4-literal	3	130	b	3.4×10^{-14}
c5	1	20	value	2.2×10^{-19}
c5-fast	3	28	CI	7.1×10^{-15}
c5-literal	3	28	CI	5.2×10^{-15}
c6	1	110	value	5.4×10^{-20}
c6-fast	3	130	CI	5.9×10^{-13}
c6-literal	3	130	CI	3.5×10^{-13}
c7	1	110	value	6.9×10^{-18}
c7-fast	3	130	CI	3.7×10^{-13}
c7-literal	3	130	CI	1.6×10^{-13}
c8	1	110	value	2.2×10^{-19}
c8-fast	3	130	CI	4.3×10^{-14}
c8-literal	3	130	CI	3.4×10^{-14}

xthenreg: complete result

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
c1	6	266	CI	3.1×10^{-13}
c2	6	114	CI	2.2×10^{-14}
c3	6	182	CI	1.4×10^{-12}
c4	6	133	CI	4.3×10^{-14}
c5	6	62	CI	7.1×10^{-15}
c6	6	133	CI	5.9×10^{-13}
c7	6	133	CI	3.7×10^{-13}
c8	6	133	CI	4.3×10^{-14}
gb1	6	31	CI	1.1×10^{-14}

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
gb2	6	57	CI	1.2×10^{-14}
gb3	6	31	V	4.1×10^{-11}
gb4	6	21	CI	8.7×10^{-14}
gb5	6	31	V	2.7×10^{-11}
gk1	6	43	CI	7.8×10^{-15}
gk2	6	31	CI	4.9×10^{-15}
gk3	6	43	CI	1.1×10^{-14}
gt1	6	133	CI	6.7×10^{-12}
gt2	6	57	CI	1.2×10^{-12}
gt3	6	57	CI	5.7×10^{-14}
gt4	6	133	CI	9.3×10^{-13}
gv01	6	91	CI	1.2×10^{-13}
gv02	6	91	CI	1.4×10^{-14}
gv03	6	91	CI	1.9×10^{-14}
gv04	6	91	CI	8.5×10^{-14}
gv05	6	91	CI	2.8×10^{-14}
gv06	6	91	CI	2.1×10^{-14}
gv07	6	91	CI	1.4×10^{-13}
gv08	6	91	CI	7.3×10^{-15}
gv09	6	91	CI	2.6×10^{-14}
gv10	6	91	CI	1.7×10^{-13}
gv11	6	91	CI	2.1×10^{-14}
gv12	6	91	CI	3.7×10^{-14}

xthenreg: bandwidth multiplier h_0() through the public function

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
c7	3	130	CI	3.7×10^{-13}
c8	3	130	CI	4.3×10^{-14}

xtdpthresh: instruments and fixed-threshold GMM

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
collapse	1	11 200	value	1.9×10^{-16}
common_variable	1	29 400	value	1.9×10^{-16}
common_variable_collapse	1	7 000	value	1.9×10^{-16}
dependent_iv	1	49 000	value	1.9×10^{-16}
duplicate_endog_lag	1	49 000	value	1.9×10^{-16}
duplicate_predetermined_ylag	1	17 500	value	1.7×10^{-16}
endog	1	49 000	value	1.9×10^{-16}
endog_predet	1	68 600	value	0
exog_only	1	29 400	value	1.9×10^{-16}
exogenous_option	1	34 300	value	6.7×10^{-16}
iv_one	1	34 300	value	4.4×10^{-16}
iv_two	1	58 800	value	5.1×10^{-15}
iv_two_collapse	1	50 400	value	5.1×10^{-15}
iv_with_collapse	1	7 000	value	4.4×10^{-16}
kink	1	49 000	value	1.9×10^{-16}
maxlag_1_2	1	24 500	value	0
maxlag_2_5	1	40 600	value	1.9×10^{-16}
maxlag_4_4	1	16 800	value	1.9×10^{-16}
maxlag_collapse	1	7 000	value	1.9×10^{-16}
nocenter	1	49 000	value	1.9×10^{-16}
predet	1	53 900	value	1.9×10^{-16}
static	1	14 700	value	8.9×10^{-16}
static_endog	1	35 200	value	1.9×10^{-16}
unbalanced_alternate_years	1	34 300	value	1.9×10^{-16}
unbalanced_gaps	1	35 560	value	1.9×10^{-16}
unbalanced_missing_endog	1	42 000	value	1.9×10^{-16}
unbalanced_missing_x	1	26 040	value	1.1×10^{-16}

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
unbalanced_short_units	1	39 200	value	1.9×10^{-16}

xtdpthresh: threshold search and final estimate

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
alternate_years_grid10	6	61	e(V_cond)	3.4×10^{-14}
alternate_years_kink	6	25	e(b)	3.8×10^{-15}
collapse	6	61	e(V_cond)	5.2×10^{-13}
default	6	61	e(obj)	1.5×10^{-11}
exog_only	6	35	e(obj)	2.1×10^{-12}
grid10	6	61	e(obj)	6.3×10^{-11}
grid200	6	61	e(obj)	1.1×10^{-10}
iv	6	61	e(obj)	4.0×10^{-11}
kink	6	25	e(obj)	3.8×10^{-10}
kink_grid10	6	25	e(obj)	2.4×10^{-10}
kink_trim20	6	25	e(obj)	7.7×10^{-12}
min_regime	6	61	e(obj)	6.6×10^{-11}
nocenter	6	61	e(obj)	8.7×10^{-12}
observed_support	6	61	e(obj)	4.2×10^{-11}
quantile	6	61	e(obj)	1.0×10^{-10}
quantile_10	6	61	e(obj)	4.3×10^{-11}
refine1	6	61	e(obj)	2.4×10^{-12}
refine3	6	61	e(obj)	2.4×10^{-12}
refine_quantile	6	61	e(obj)	7.2×10^{-11}
static	6	35	e(obj)	2.0×10^{-13}
trim05	6	61	e(obj)	1.2×10^{-10}
trim20	6	61	e(obj)	7.7×10^{-11}
unbalanced_alternate_years	6	61	e(b)	1.9×10^{-14}
unbalanced_gaps	6	61	e(obj)	1.1×10^{-11}
unbalanced_kink	6	25	e(obj)	9.5×10^{-10}
unbalanced_missing_endog	6	61	e(obj)	3.8×10^{-11}
unbalanced_missing_x	6	61	e(obj)	1.6×10^{-11}
unbalanced_refine	6	61	e(obj)	1.8×10^{-11}
unbalanced_short_units	6	61	e(obj)	4.7×10^{-10}

xtdpthresh: point estimates on the full panel

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
default_instruments	4	58	e(obj)	1.6×10^{-10}
help_grid_200	4	92	e(obj)	3.3×10^{-13}
help_iv_peer	4	92	e(obj)	6.0×10^{-13}
help_kink	4	32	e(obj)	1.1×10^{-12}
help_maxlag_2_10	4	92	e(obj)	1.1×10^{-9}
help_maxlag_3_8	4	92	e(obj)	2.9×10^{-13}
help_quantile_grid	4	92	e(obj)	1.4×10^{-13}
help_refine	4	92	e(obj)	2.7×10^{-13}
help_unbalanced_alternate	4	92	e(obj)	1.1×10^{-12}
help_unbalanced_gaps	4	92	e(obj)	1.8×10^{-13}
help_unbalanced_short	4	92	e(obj)	7.3×10^{-14}
kink_default_instruments	4	22	e(obj)	4.5×10^{-10}
mixed_roles	4	92	e(b)	1.9×10^{-14}
static_default	4	32	e(obj)	5.0×10^{-13}

xtdpthresh: deviations, time effects, unbalanced panels, history

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
fd_age_without_td	7	55 403	e(b)	1.9×10^{-10}
fd_history_panel	7	33 567	e(obj)	4.7×10^{-12}

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
fd_history_sample	7	14 067	e(obj)	3.6×10^{-13}
fd_iv_with_gaps	7	42 739	e(obj)	1.7×10^{-10}
fd_td_alternate_years	7	34 367	e(obj)	2.0×10^{-10}
fd_td_collapse	7	15 467	e(obj)	6.9×10^{-15}
fd_td_endog	7	49 067	e(obj)	1.6×10^{-11}
fd_td_exog_only	7	28 039	e(obj)	1.8×10^{-13}
fd_td_gaps	7	35 627	e(obj)	1.2×10^{-11}
fd_td_kink	7	46 928	e(obj)	4.4×10^{-11}
fd_td_short_units	7	39 267	e(obj)	6.4×10^{-11}
fd_td_single_period	7	33 067	e(obj)	1.1×10^{-11}
fd_td_sparse_periods	7	14 036	e(b)	9.0×10^{-15}
fod_alternate_years	7	37 797	e(b)	8.2×10^{-15}
fod_collapse	7	16 167	e(V_cond)	3.1×10^{-14}
fod_endog	7	53 967	e(obj)	3.0×10^{-11}
fod_exog_only	7	32 939	e(obj)	2.2×10^{-12}
fod_gaps	7	42 725	e(obj)	1.8×10^{-11}
fod_history_panel	7	33 567	e(obj)	4.9×10^{-11}
fod_history_sample	7	14 067	e(obj)	3.1×10^{-13}
fod_iv	7	63 767	e(obj)	3.1×10^{-10}
fod_iv_with_gaps	7	46 603	e(obj)	5.2×10^{-11}
fod_kink	7	51 828	e(obj)	4.5×10^{-11}
fod_maxlag	7	45 567	e(obj)	9.2×10^{-12}
fod_missing_endog	7	50 117	e(obj)	2.0×10^{-11}
fod_missing_x	7	38 569	e(obj)	6.8×10^{-11}
fod_nocenter	7	53 967	e(obj)	4.0×10^{-12}
fod_predet	7	58 867	e(obj)	1.2×10^{-11}
fod_refine	7	53 967	e(obj)	1.5×10^{-11}
fod_short_units	7	43 187	e(obj)	6.4×10^{-11}
fod_sparse_periods	7	26 701	e(V_cond)	4.7×10^{-14}
fod_static	7	18 239	e(obj)	6.4×10^{-14}
fod_td_alternate_years	7	34 367	e(obj)	4.1×10^{-11}
fod_td_collapse	7	15 467	e(obj)	3.0×10^{-14}
fod_td_endog	7	49 067	e(obj)	6.9×10^{-12}
fod_td_exog_only	7	28 039	e(obj)	6.4×10^{-13}
fod_td_gaps	7	42 725	e(obj)	4.2×10^{-12}
fod_td_short_units	7	39 267	e(obj)	1.6×10^{-11}
fod_td_single_period	7	36 728	e(obj)	7.6×10^{-9}
fod_td_sparse_periods	7	25 543	e(b)	2.7×10^{-13}
fod_td_static	7	13 339	e(obj)	2.6×10^{-14}

xtdpthresh: variances and diagnostics

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
fd_alternate_wind	29	156	e(ar2_cond)	3.4×10^{-14}
fd_bw_large	32	162	e(hansen)	1.7×10^{-11}
fd_bw_small	32	162	e(hansen)	1.7×10^{-11}
fd_collapse_wind	32	162	e(V)	6.1×10^{-14}
fd_discrete_q	32	162	e(hansen)	4.5×10^{-10}
fd_exog_only	32	106	e(hansen)	1.6×10^{-12}
fd_exog_only_wind	32	106	e(hansen)	1.6×10^{-12}
fd_gaps	32	162	e(hansen)	1.6×10^{-11}
fd_gaps_wind	32	162	e(hansen)	1.6×10^{-11}
fd_iv_wind	32	162	e(hansen)	1.3×10^{-10}
fd_jump	32	162	e(hansen)	1.7×10^{-11}
fd_jump_q_regressor	32	236	e(hansen)	1.1×10^{-9}
fd_jump_wind	32	162	e(hansen)	1.7×10^{-11}
fd_kink	28	77	e(hansen)	3.7×10^{-10}
fd_kink_q_regressor	30	98	e(hansen)	6.8×10^{-10}
fd_kink_wind	28	77	e(hansen)	3.7×10^{-10}
fd_nocenter	32	162	e(hansen)	1.8×10^{-12}
fd_nocenter_wind	32	162	e(hansen)	1.8×10^{-12}
fd_predet	32	162	e(hansen)	2.2×10^{-10}
fd_refine_wind	32	162	e(hansen)	5.1×10^{-12}

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
fd_short_wind	32	162	e(hansen)	4.6×10^{-10}
fd_static	32	106	e(hansen)	2.0×10^{-13}
fd_td	32	162	e(hansen)	8.0×10^{-12}
fd_td_wind	32	162	e(hansen)	8.0×10^{-12}
fod_alternate_wind	29	156	e(ar1_cond)	9.8×10^{-15}
fod_bw_small	32	162	e(hansen)	1.0×10^{-11}
fod_gaps_wind	32	162	e(hansen)	4.5×10^{-12}
fod_jump	32	162	e(hansen)	1.0×10^{-11}
fod_jump_wind	32	162	e(hansen)	1.0×10^{-11}
fod_kink	28	77	e(hansen)	2.7×10^{-11}
fod_kink_wind	28	77	e(hansen)	2.7×10^{-11}
fod_nocenter_wind	32	162	e(hansen)	7.4×10^{-13}
fod_short	32	162	e(hansen)	1.4×10^{-10}
fod_td	32	162	e(hansen)	2.6×10^{-12}
fod_td_wind	32	162	e(hansen)	2.6×10^{-12}
full_help	32	236	e(ar1_TT)	3.7×10^{-12}
full_help_fod	32	236	e(ar1_TT)	7.3×10^{-13}
full_help_fod_wind	32	236	e(ar1_T1)	5.9×10^{-13}
full_help_kink	30	98	e(hansen)	1.1×10^{-12}
full_help_kink_wind	30	98	e(hansen)	1.1×10^{-12}
full_help_wind	32	236	e(ar2_TT)	3.9×10^{-13}
full_maxlag_2_10_wind	32	236	e(hansen)	1.1×10^{-9}
full_unbalanced_alternate_wind	32	236	e(ar1_T1)	2.6×10^{-12}
full_unbalanced_gaps_wind	32	236	e(ar1_b0)	2.5×10^{-13}
full_unbalanced_short	32	236	e(ar1_TT)	8.7×10^{-13}

xtdpthresh: predict

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
fd_jump	5	3 500	predict residuals	4.3×10^{-14}
fd_kink	5	3 500	predict residuals	2.5×10^{-13}
fd_short_wind	5	2 830	predict residuals	3.3×10^{-13}
fd_td	5	3 500	predict residuals	1.3×10^{-14}
fod_alternate_wind	5	2 530	predict ar_residuals	1.7×10^{-15}
fod_gaps_wind	5	2 743	predict ar_residuals	1.8×10^{-14}
fod_jump	5	3 400	predict ar_residuals	2.7×10^{-14}
fod_kink	5	3 400	predict ar_residuals	8.8×10^{-15}
fod_short	5	2 750	predict ar_residuals	2.8×10^{-13}
fod_td	5	3 400	predict ar_residuals	2.3×10^{-14}
full_help_fod	5	34 465	predict ar_residuals	8.6×10^{-14}
xb_plus_residuals_is_the_transformed_dependent_variable	1	700	xb + residuals	6.9×10^{-18}

xtdpthresh: bootstraps, confidence set, tests

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
citest_fod	20	272	e(ci_grid)	2.8×10^{-9}
citest_inside	20	270	e(ci_grid)	1.1×10^{-9}
citest_kink	20	270	e(ci_grid)	4.3×10^{-10}
citest_one_step	20	270	e(ci_grid)	5.6×10^{-16}
citest_outside	20	270	e(citest_crit)	1.4×10^{-9}
citest_unseeded_subset	20	270	e(ci_grid)	1.1×10^{-9}
citest_with_tests	22	278	e(ci_grid)	9.9×10^{-9}
coef_onestep	20	283	e(ci_grid)	1.1×10^{-9}
coef_onestep_percentile	20	283	e(ci_grid)	1.1×10^{-9}
coef_onestep_td	20	287	e(ci_grid)	3.1×10^{-10}
coef_twostep	20	283	e(ci_grid)	1.1×10^{-9}
coef_twostep_collapse	20	287	e(ci_grid)	1.6×10^{-12}
coef_twostep_fod	20	285	e(ci_grid)	2.8×10^{-9}

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
coef_twostep_kink	20	277	e(ci_grid)	4.3×10^{-10}
coef_twostep_one_step_fit	20	283	e(b_bootci)	3.1×10^{-14}
coef_twostep_percentile	20	283	e(ci_grid)	1.1×10^{-9}
coef_twostep_refine	20	293	e(ci_grid)	1.5×10^{-9}
cont_fd	15	271	e(ci_grid)	9.9×10^{-9}
cont_fd_collapse	15	255	e(ci_grid)	8.2×10^{-10}
cont_fd_gaps	15	269	e(ci_grid)	3.8×10^{-8}
cont_fd_one_step	15	259	e(ci_grid)	3.1×10^{-15}
cont_fd_static	15	269	e(ci_grid)	1.3×10^{-10}
cont_fd_td	15	265	e(ci_grid)	2.9×10^{-9}
cont_fod	15	267	e(ci_grid)	4.0×10^{-9}
fd_collapse	13	267	e(ci_grid)	1.6×10^{-12}
fd_exog_only	13	259	e(ci_grid)	2.0×10^{-11}
fd_gaps	13	267	e(ci_grid)	2.4×10^{-9}
fd_iv	13	265	e(ci_grid)	4.0×10^{-8}
fd_jump	13	263	e(ci_grid)	1.1×10^{-9}
fd_kink	13	263	e(ci_grid)	4.3×10^{-10}
fd_level_90	13	263	e(ci_grid)	1.5×10^{-9}
fd_no_linearity_test	11	260	e(ci_grid)	1.1×10^{-9}
fd_one_step	13	263	e(ci_grid)	5.6×10^{-16}
fd_quantile_grid	13	267	e(ci_grid)	2.9×10^{-9}
fd_refine	13	273	e(ci_grid)	1.5×10^{-9}
fd_short	13	267	e(ci_grid)	8.9×10^{-8}
fd_small_boot	13	271	e(ci_grid)	9.3×10^{-10}
fd_static	13	273	e(ci_grid)	1.6×10^{-12}
fd_td	13	267	e(ci_grid)	3.1×10^{-10}
fod_jump	13	265	e(ci_grid)	2.8×10^{-9}
fod_kink	13	261	e(ci_grid)	2.0×10^{-10}
fod_one_step	13	269	e(ci_grid)	1.2×10^{-15}
fod_td	13	267	e(ci_grid)	2.9×10^{-10}
full_citest	20	266	e(ci_grid)	2.4×10^{-10}
full_cont	15	261	e(ci_grid)	2.4×10^{-10}
full_help	13	259	e(ci_grid)	2.4×10^{-10}
full_help_fod	13	265	e(ci_grid)	1.1×10^{-9}
full_help_kink	13	257	e(ci_grid)	1.6×10^{-11}
unit_citest	11	138	e(ci_grid)	5.0×10^{-16}
unit_collapse	13	263	e(ci_grid)	6.3×10^{-12}
unit_exog_only	13	257	e(ci_grid)	1.9×10^{-9}
unit_fd	8	135	e(ci_grid)	5.0×10^{-16}
unit_gaps	8	130	e(ci_grid)	4.4×10^{-16}
unit_nocenter	13	263	e(ci_grid)	3.7×10^{-12}
unit_quantile_grid	8	134	e(ci_grid)	4.7×10^{-16}
unit_static	13	261	e(ci_grid)	8.0×10^{-12}

xtdpthresh: unbalanced panel of the help file (no bootstrap)

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
fd	4	92	e(obj)	6.6×10^{-14}
fod	4	92	e(obj)	5.0×10^{-14}

xtdpthresh: hostile inputs

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
iv_equals_x	1	5	value	4.8×10^{-8}
q_four_values	1	7	value	4.9×10^{-15}
units_fewer_than_iv	1	5	value	9.2×10^{-9}
units_fewer_than_iv_boot	1	7	value	5.9×10^{-8}

Simulated panels of the Seo-Shin design

Case	Objects	Values	Worst object	$\lvert\Delta\rvert$
1	1	5	value	3.4×10^{-9}
2	1	5	value	4.5×10^{-9}
3	1	5	value	4.5×10^{-9}
4	1	5	value	4.3×10^{-9}

6 Comparisons under a looser tolerance

The rule of Section 1 holds for every other value of this report. 4 comparison(s) are made under a looser relative tolerance, declared in the test and not hidden in the tables above: the input is so ill-conditioned that the error of any implementation is of the order of the condition number times the machine precision (a matrix of condition number 10^{13} gives about 10^{-3}), so two correct inverses legitimately differ in the fourth digit.

Case	Object	Relative tolerance	$\lvert\Delta\rvert$
cond13	invsym	1.0×10^{-3}	6.0×10^{1}
cond13	cholinv	1.0×10^{-3}	2.3×10^{8}
cond13	syminv_ok	1.0×10^{-3}	0
cond13	psd	1.0×10^{-3}	0

7 What this report does not cover

- The statistical behaviour of the package's own seeds (size and power of the tests, coverage of the confidence set over many panels) is measured by Monte Carlo and compared with the papers, not with Stata.
- The tables written by the package (LaTeX, HTML, Word, Excel) repeat the validated numbers; the tests check the printed digits, not the typography. They are not in the deviations above.
- Timings are measured on the same machine one run after the other and are not a validation.
- The messages that Stata prints and the package does not (the warning of an unbalanced `boottype(unit)` and the hint on near-common variables in a rank failure) are not compared; they do not change any number.

8 Environment

Reference	Stata 17.0 MP; xthenreg (Seo, Kim and Kim 2019) and xtdpthresh 0.9.38
Package	pyxtdpthresh 0.0.1
Python	3.12.10; NumPy 2.5.4; Numba 0.68.1
Platform	Windows 11
Report built	03 November 2026